Review: *The worker/harness weld is the regular bicomodule: deciding Conjecture 5.1*

Rick

To: Scot MacBeth (Kodamai)

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Reviewed:	MacBeth WIP `2fd555a` (5 pp., email UID 405, 2026-10-17)
This review:	corresponds to clio-review commit `3c848d1`; reviewing MacBeth WIP `2fd555a`
Check script:	`2026-10-17-macbeth-uid405-weld-check.py` (+ `.log`), brute force, <1 s

Notation. I write $\mathsf{St}_X := Xy^X$ for your ΔX (store comonoid = indiscrete category on X), because you also use Δ_{TS} for a diagonal in §4. All sets nonempty.

Verdict

- **Clause (1) and Theorem 5: correct.** I re-derived both by hand and by a direct Hom-count in St_T-**Set** (not via your factorization through y): $\bar{F}_B(V) \cong T^S \times V^S$ at all seven shapes run, including the generic $(2,3)$ and $(3,2)$.
- **(a) Handedness.** Literal answer: yes, no handedness choice with a nontrivial right coaction gives exponent $S \times T$. The transported exponent at a position is the number of right-coaction orbits on $B[i] = S \times T$: $|S|$ (right St_T, yours), $|T|$ (right St_S, or $\triangleleft$ swapped), 1 (right $\mathsf{St}_{S\times T}$). The exponent is $|S \times T|$ *exactly* when the right comonoid is y, i.e. no right coaction. **But that is the reading you used for** A (Thm 6 proof: $\bar{A}(V) = V^{S\times T}$ is A as a left $\mathsf{St}_{S\times T}$-comodule and nothing else). So Theorem 6 compares B under one reading with A under another. Under any *consistent* reading the exponents agree. The refutation as written fails, and it fails at a point you list: at $|S| = 1$, $|T| \geq 2$ we have $\mathsf{St}_S = y$ and $B = y \triangleleft \mathsf{St}_T = \mathsf{St}_T = A$ *on the nose*, yet Theorem 6 says the worker is not the content. The argument proves too much. **Verdict FALSE survives, for a different reason, on a different locus** (M1).
- **(b) σ.** $\sigma_1(s,t) = (s, \mathrm{const}_t)$ is well defined, injective ($S \neq \emptyset$), a section of $r(s,g) = (s, g(s))$, and its image is closed under λ and ρ, so $\sigma(A) \subseteq B$ is a sub-bicomodule. Yes, $T \subseteq T^S$ means constant maps (diagonal). But: σ is not a comonoid map (no comonoid on B is in play), the structure $\sigma(A)$ inherits (left St_S, right St_T) is not A's own $\mathsf{St}_{S\times T}$, the direction map $\sigma^\sharp$ is never stated, and the §5 display mislocates the S (M3).

What the paper should say for clause (2)

Use $\sigma(A)$ as the sub-bicomodule. Its prafunctor is $\bar{F}_{\sigma A}(V) = T \times V^S$; that of B is $T^S \times V^S$. Same exponent; Σ-index T versus T^S. Since $|S \times T| = |S \times T^S|$ iff $\min(|S|, |T|) = 1$, the worker is all of the

weld iff $|S| = 1$ or $|T| = 1$, and is the proper constant-branch slice of the Σ-tail otherwise. That is a clean refutation for $|S|, |T| \geq 2$ and it is your §5 idea promoted to the proof. Script line "clause (2)": your criterion $|T| \geq 2$ and this one disagree at $(1, 2)$ and $(1, 3)$, agree elsewhere.

Findings

MUST

M1. p. 4, Theorem 6 and proof; sentence "No handedness convention repairs this." Inconsistent comparison (above). Replace the exponent argument by the Σ-index argument; change the failure locus from "$|T| \geq 2$" to "$|S| \geq 2$ and $|T| \geq 2$"; state which reading of "worker = content" is being refuted (clause (2) is not type-correct as written: A is a polynomial/comonoid, $F/1$ is a functor $\mathbf{Set} \to \mathbf{Set}^{T^S}$). The "opposite reading gives $(-)^T$" remark is right but irrelevant to the gap.

M2. p. 4, Theorem 7. "indexed by the *full* branch-map set T^S — precisely the non-constant branch maps" is self-contradictory: at $(2, 3)$, $|T^S| = 9$, non-constant $= 6$; at $|S| = 1$ nothing is discarded but the index is T. Per position, r forgets $g|_{S \setminus \{s\}}$ (fibre $T^{|S|-1}$: 3 at $(2, 3)$, 4 at $(3, 2)$), not "the non-constant maps". True statement: $T^S = \mathrm{const}(T) \sqcup$ (non-constant), kept slice $\sqcup$ discarded part. Clause (3) as stated (discarded = Σ-tail) is false; restate as "Σ-tail index = kept slice $\sqcup$ discarded data" and re-decide.

M3. p. 5, §5 display. Two different S's are conflated. A's position coordinate s is the St_S-object (the left-coaction fibre, which disappears under St_S-$\mathbf{Set} \simeq \mathbf{Set}$). The content exponent S is the first *direction* coordinate a of $(a, t) \in S \times T$. σ sends position s to position s; it puts nothing into the exponent. Replace with: $\sigma(A)$ is the sub-bicomodule on positions $\{(s, \mathrm{const}_t)\}$ (closed since $\lambda(s, g)$ moves s and keeps g, and ρ fixes positions), $\sigma^\sharp = \mathrm{id}_{S \times T}$, prafunctor $T \times V^S$.

SHOULD

S1. p. 1, Provenance and every "SGF" locator. arXiv:2111.10968 is Spivak–Garner–Fairbanks, *Functorial aggregation*. The title you give is Niu–Spivak, arXiv:2312.00990. So the source of Def. 4.2, Rmk. 4.3, Thm. 4.28, Props. 3.6, 3.9, 5.2, 5.7, Ex. 5.6 is ambiguous. I checked none of them. Fix the bibliography and re-check every locator against the print.

S2. p. 2, definition of a (c, d)-bicomodule. As I read Niu–Spivak, a bicomodule *crightarrowd* has a left c- and a right d-coaction and gives d-$\mathbf{Set} \to c$-$\mathbf{Set}$. Your labels are the other way. You are internally consistent, so nothing breaks, but say explicitly that the labels are reversed relative to the source. Also, "regular bicomodule" usually means c as a (c, c)-bicomodule. $d \triangleleft c$ is the composite over y of d and c (your own Thm 5 proof says so). Rename it, and the title with it.

S3. p. 3, machine checks. The Theorem 5 corroboration (16/64/128) is at $|S| = |T| = 2$. That cannot separate $T^S y^S$ from the swapped $S^T y^T$: both give 16 at $V = 2$ and 36 at $V = 3$ (script). At $(2, 3)$ they give 36 vs 64. The axiom check is on Lemma 3, which is general nonsense, and includes the degenerate shapes $(1, 2)$, $(2, 1)$. So it corroborates nothing in dispute. Check $\bar{F}_B$ by a direct Hom-count at a generic shape.

S4. p. 1, registry. Node `weld-is-regular-bicomodule` [proved] bundles (1)–(3). (1) stands. (2) and (3) do not, as written. Split the node; the bundle inherits its weakest input.

S5. pp. 3–4. Put the nonemptiness hypothesis in the statements, not in a parenthesis. St_S-**Set** $\simeq$ **Set** and the injectivity of $T \hookrightarrow T^S$ both need $S \neq \emptyset$.

NIT

N1. Δ means both store comonoid and diagonal Δ_{TS} (§4). Rename one.

N2. p. 3, Thm 5 proof: "no equalizer, since y is the unit" — say "the composite over y is plain $\triangleleft$".

N3. p. 5, §6: "read the whole input state S" is misleading. After transport, $\bar{F}_B$ is an endofunctor of **Set**. The state lives only in the fibration over St_S-objects and in the direction coordinate a. Drop the gloss or make it precise.

Grade

Clause (1) and Theorem 5: **checked-sober** (re-derived line by line, independent Hom-count). Clause (2): verdict FALSE **checked-sober** only in the corrected form (Σ-index, $|S|, |T| \geq 2$). Your argument and locus are **not accepted**. Clause (3): false as stated; needs restating. Overall claim as submitted: **sketched**.

Appendix: script output (abridged)

```
(|S|,|T|)  exp: y  St_T  St_S  St_SxT   |T^S|  noncst  A=B?   (2) yours / consistent
 (1,2)          2   1     2     1        2      0      True    FALSE / holds   DISAGREE
 (2,1)          2   2     1     1        1      0      True    holds / holds
 (2,2)          4   2     2     1        4      2      False   FALSE / FALSE
 (2,3)          6   2     3     1        9      6      False   FALSE / FALSE
 (3,2)          6   3     2     1        8      6      False   FALSE / FALSE
F_B(V) at (2,2): 16, 36 = swapped order 16, 36  [no separation]
F_B(V) at (2,3): 36, 81 vs swapped 64, 216
sigma: injective, r.sigma = id, image closed under lambda,rho: True at all shapes
```